

Feb 23, 2024

Introduction to Computer and Network Security

Part 1

Question 1 [60 points]

Energy management and automation giant AiElectric suffered a ransomware attack leading to the theft of corporate data and its subsequent encryption in the company storage. The ransomware gang reportedly exploited a decade-old vulnerability (for which exists a fix), stole terabytes of corporate data during the cyberattack and is now extorting the company by threatening to leak the stolen data if a ransom demand is not paid.

You are a security expert paid to comment on why it is *virtually* impossible to decrypt AiElectric's data without paying the ransom. You are inherently able to do it by describing the notions of

- (a) cryptosystem [5 points];
- (b) Kerckhoffs principle [5 points];
- (c) key management and its purpose [10 points];
- (d) symmetric key cryptography [10 points];
- (e) asymmetric key cryptography [10 points];
- (f) what is the difference between a hash and an encryption function [10 points].

Finally, you should describe

- (g) which security architecture and best practices could have mitigated this kind of attack [10 points].

Part 2

Question 2 [30 points]

In the context of Zero Trust, describe (a) the concept of zero trust and its six pillars and (b) the concept of access control.

Question 3 [10 points]

Briefly describe OAuth 2.0 and discuss whether authenticating with it is a good or bad idea.

Please, remember...

Be sure to annotate the following information on each used sheet of paper (possibly, 2 sheets at most)

- Name
- Surname
- Student ID number
- Question number